

Nettrix 宁畅		制定日期	2023/11/05			
		修改日期				
拟制部门	核 准	关联部门		制 定 人	版 本	总 页 数
研发中心	胡远明			范一博	V1.0	共 35 页

Nettrix 宁畅

变更记录

[illegible]

LeCroy PCIe协议分析软件使用介绍

宁畅信息产业（北京）有限公司

界面介绍

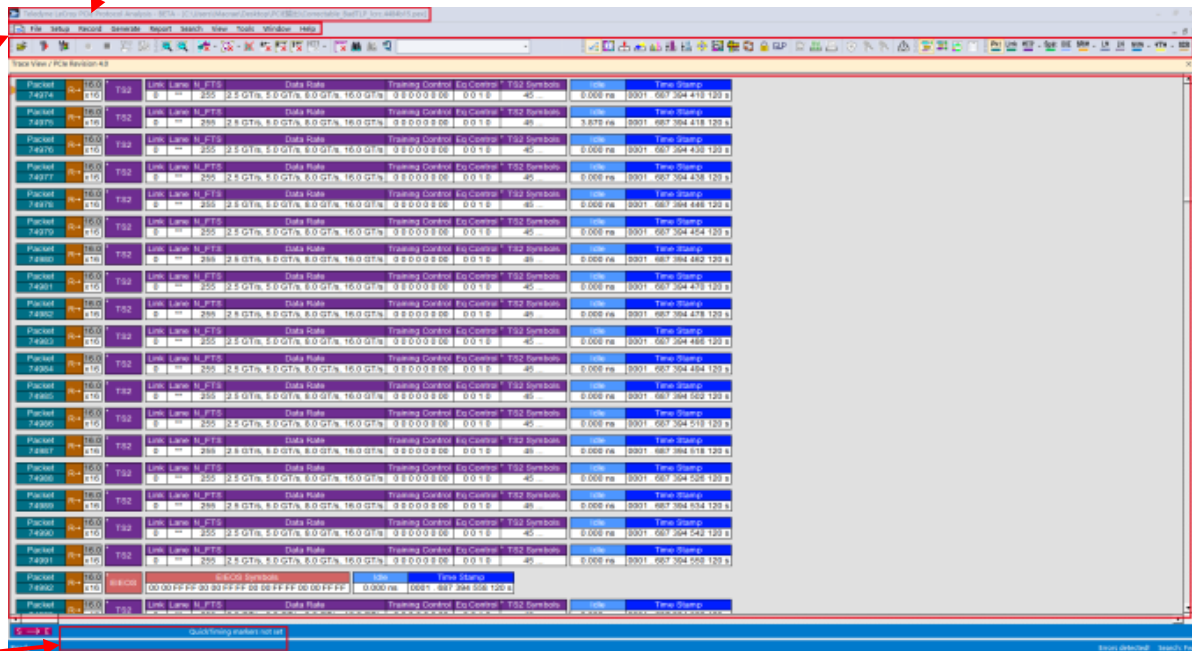
标题
栏

菜单
栏

工具
栏

显示区
域

状态栏：连接
仪器后显示

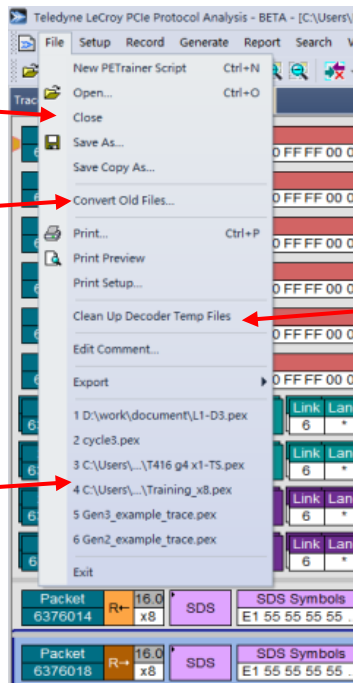


File 菜单栏

打开关闭trace
文件 (.pex)

将老版本trace
文件转为新版本
可读取文件

将老版本trace
文件转为新版本
可读取文件



清理临时文件

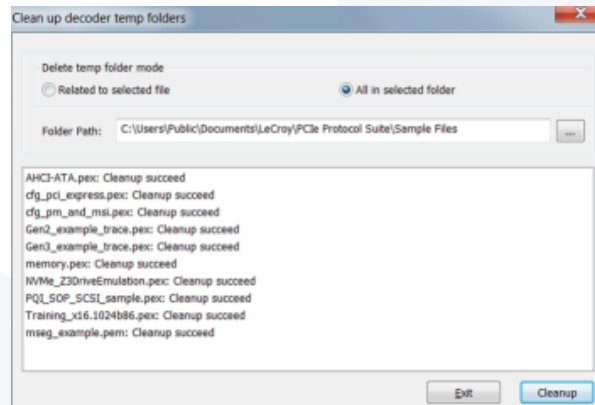
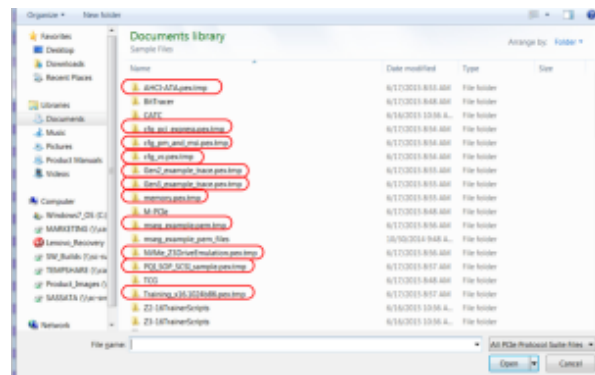
*.pem

*.pex

*.peraw

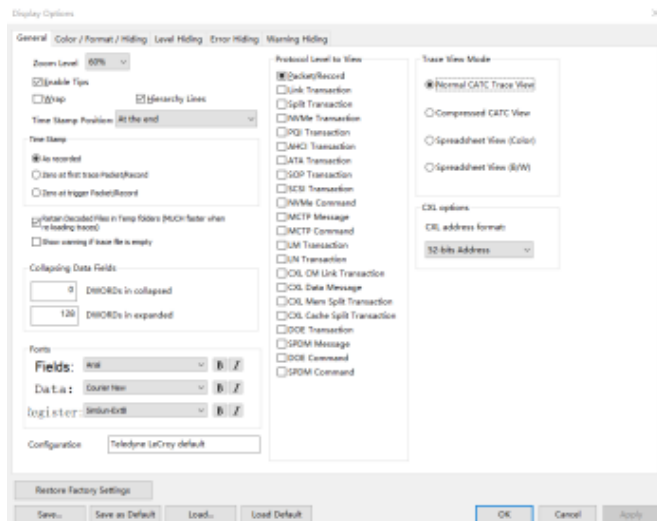
*.mpex

*.peg



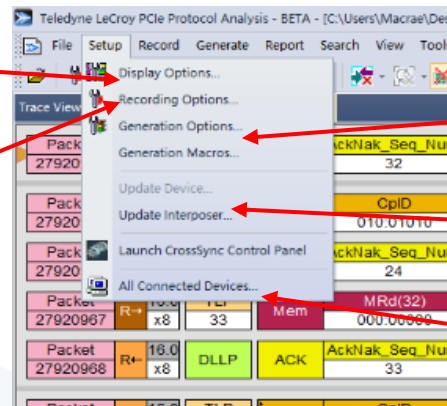
Setup 菜单栏

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显示设置包括颜色, 字体等

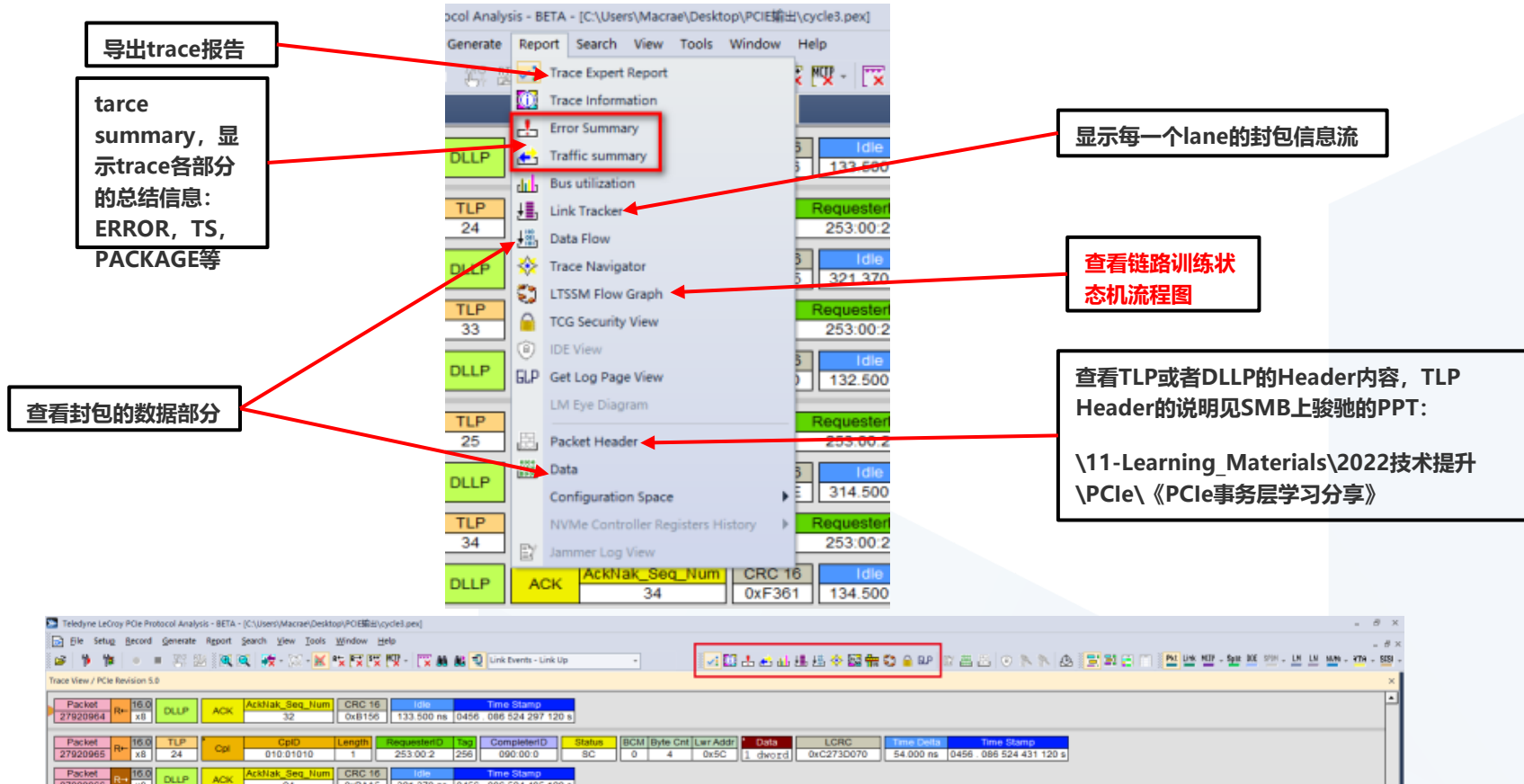
捕获trace相关设置



Z系列训练仪器
设置及宏设置

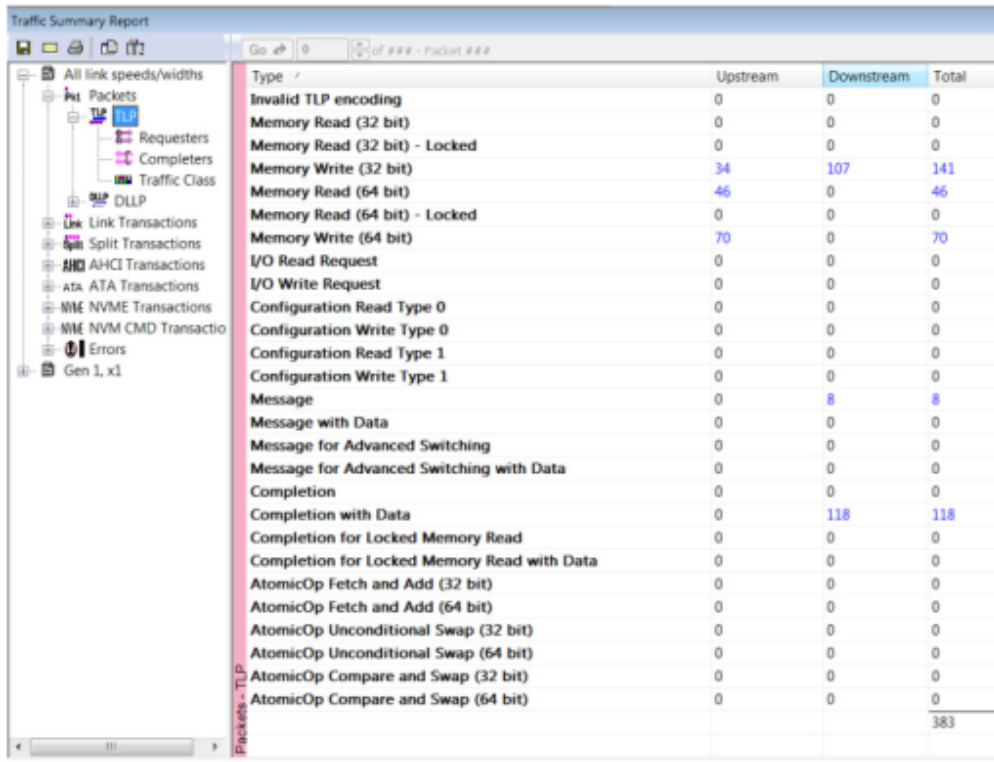
更新设备及
interposer

查看连接设备以
及通过以太网连
接控制主机



Report—Traffic Summary

使用Traffic Summary几乎能够按照分类查看Trace中的各个部分（TS, TLP, DLLP, Error等）。



The screenshot displays the 'Traffic Summary Report' window. On the left is a tree view showing the hierarchy of traffic data: All link speeds/widths, Packets, TLP (selected), Requesters, Completers, Traffic Class, DLLP, Link Transactions, Split Transactions, AHCI Transactions, ATA Transactions, NVME Transactions, NVME CMD Transactions, Errors, and Gen 1, x1. The main area on the right is a table with columns: Type, Upstream, Downstream, and Total. The table lists various transaction types and their counts. A vertical label 'Packets - TLP' is visible on the left side of the table area.

Type	Upstream	Downstream	Total
Invalid TLP encoding	0	0	0
Memory Read (32 bit)	0	0	0
Memory Read (32 bit) - Locked	0	0	0
Memory Write (32 bit)	34	107	141
Memory Read (64 bit)	46	0	46
Memory Read (64 bit) - Locked	0	0	0
Memory Write (64 bit)	70	0	70
I/O Read Request	0	0	0
I/O Write Request	0	0	0
Configuration Read Type 0	0	0	0
Configuration Write Type 0	0	0	0
Configuration Read Type 1	0	0	0
Configuration Write Type 1	0	0	0
Message	0	8	8
Message with Data	0	0	0
Message for Advanced Switching	0	0	0
Message for Advanced Switching with Data	0	0	0
Completion	0	0	0
Completion with Data	0	118	118
Completion for Locked Memory Read	0	0	0
Completion for Locked Memory Read with Data	0	0	0
AtomicOp Fetch and Add (32 bit)	0	0	0
AtomicOp Fetch and Add (64 bit)	0	0	0
AtomicOp Unconditional Swap (32 bit)	0	0	0
AtomicOp Unconditional Swap (64 bit)	0	0	0
AtomicOp Compare and Swap (32 bit)	0	0	0
AtomicOp Compare and Swap (64 bit)	0	0	0
			383

Report—Error Report

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Traffic Summary Report				
Type	Upstream	Downstream	Sideband	Total
Invalid Code	0	0	0	0
Running Disparity Error	0	0	0	0
Unexpected K/D Code	0	0	0	0
Idle Data Error (not D0.0)	1	0	0	1
Skip Late	0	0	0	0
Skew Error	0	0	0	0
Bad Packet Length	2	1	0	3
Ordered Set Format Error	2	0	0	2
Delimiter Error	0	0	0	0
Alignment Error	1	1	0	2
DLLP: Invalid Encoding	0	0	0	0
DLLP: Bad CRC16	0	0	0	0
DLLP: Reserved Field not 0	0	0	0	0
DLLP: FC Initialization Error	0	0	0	0
TLP: Invalid Encoding	0	0	0	0
TLP: Bad LCRC	0	0	0	0
TLP: Bad ECRC	0	0	0	0
TLP: Reserved Field not 0	0	0	0	0
TLP: Payload/Length Error	0	0	0	0
TLP: Length Error (not 1)	0	0	0	0
TLP: TC Error (not 0)	0	0	0	0
TLP: Attr Error (not 0)	0	0	0	0
TLP: AT Error (not 0)	0	0	0	0
TLP: Byte Enables Violation	0	0	0	0
TLP: No Header Error	0	0	0	0
Memory TLP: Address/Length Crosses 4K	0	0	0	0
Mem64 TLP: Used Incorrectly	0	0	0	0
Cfg TLP: Register Error	0	0	0	0
Msg TLP: Invalid Routing	0	0	0	0
Msg TLP: Invalid Function Number field	0	0	0	0
Gen3 TLP: Bad Len CRC/Parity	0	0	0	0
Invalid Packet	12	2	0	14
FC: Invalid Advertisement	0	0	0	0
FC: Insufficient Credits	0	0	0	0
Training Sequence Format Error	1	0	0	1
Training Sequence Parity Error	0	0	0	0
Training Sequence Cursor Error	0	210683	0	210683
Training Sequence Link Error	0	0	0	0

Report—Link Tracker

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设置封包
间隔格式

设置内容显示方式:

0x: 以16进制显示去扰解码后数据

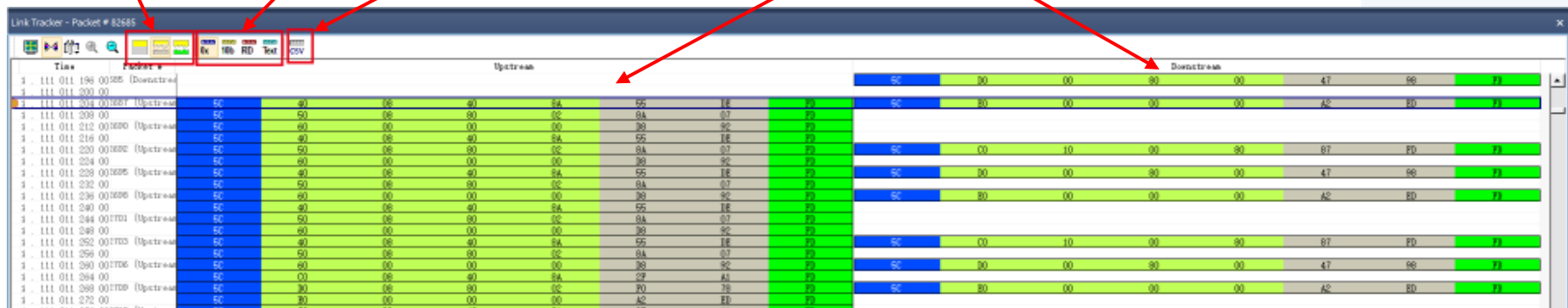
10b: 显示8b/10b编码后数据

RD: 以8b/10b信号名显示

Text: 显示该封包内容

导出成
CSV格式

不同方向
的封包信
息流

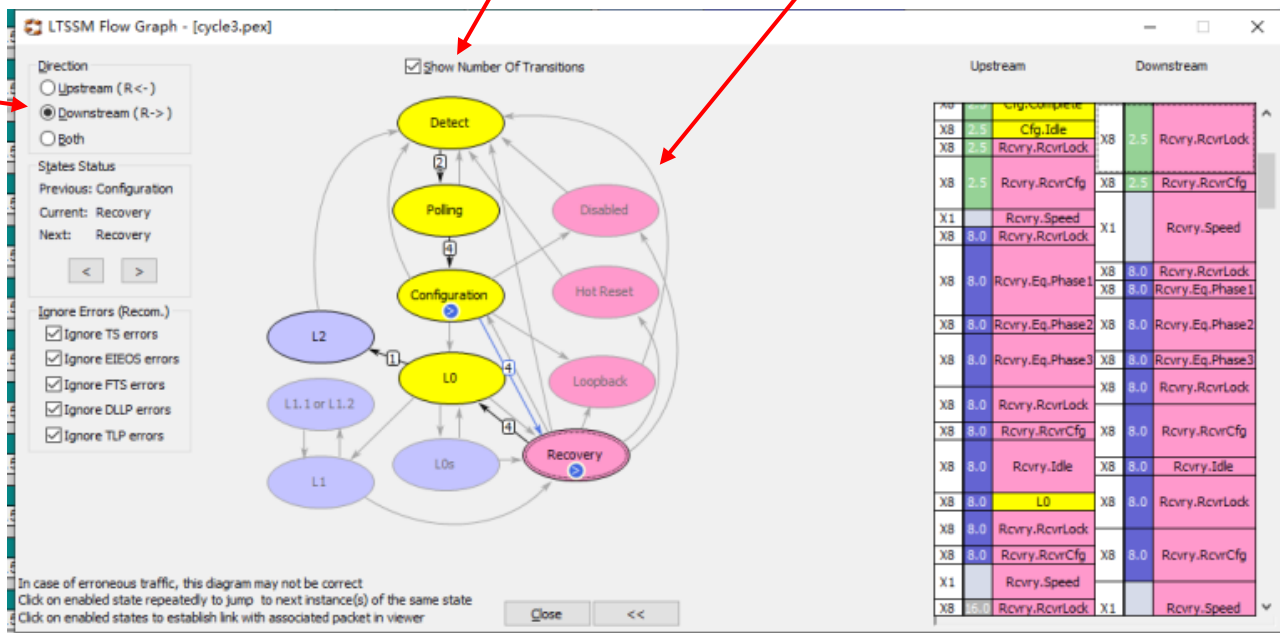


LTSSM图是我们定位问题最常用的工具之一，通过对Link Training过程分析，可以发现异常流程，超时，掉速，掉位宽等过程。

显示状态转换的次数

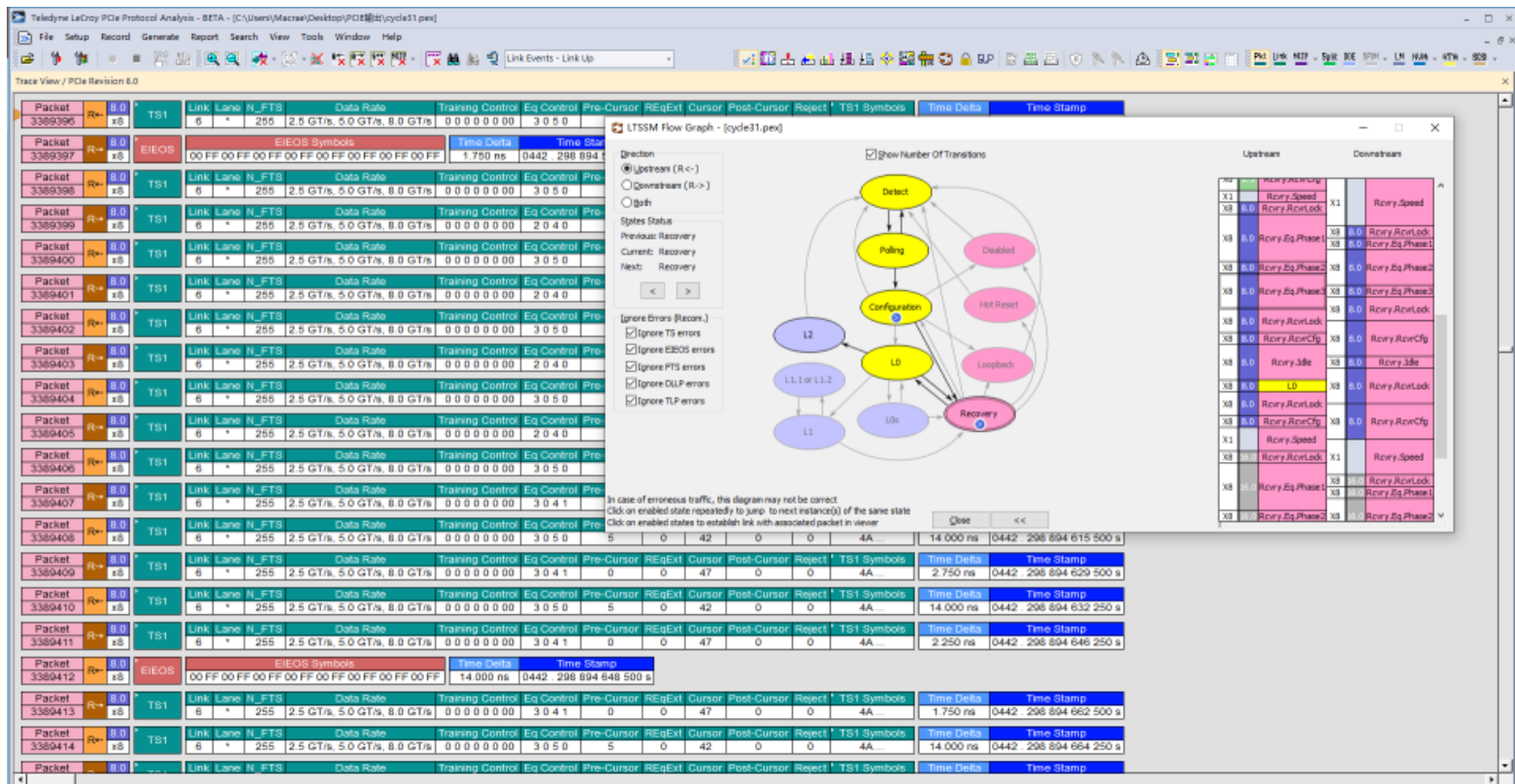
完整的LTSSM,点击箭头可以查看子状态

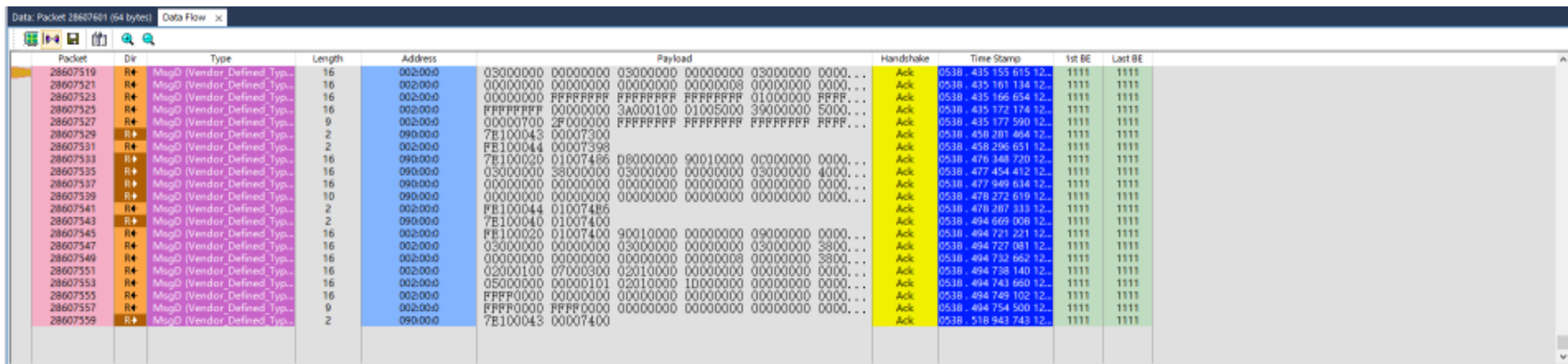
选择数据方向



8.0 GT/s以上拟合

在LTSSM中双击一个状态便可在trace中快速定位到该状态





Packet	Dir	Type	Length	Address	Payload	Handshake	Time Stamp	1st BE	Last BE
28607519	R→	MsgD (Vendor Defined Type)	16	00:00:00	03000000 00000000 03000000 00000000 03000000 0000...	Ack	0538.435.155.615.12..	1111	1111
28607521	R→	MsgD (Vendor Defined Type)	16	00:00:00	00000000 00000000 00000000 00000000 00000000 0000...	Ack	0538.435.161.134.12..	1111	1111
28607523	R→	MsgD (Vendor Defined Type)	16	00:00:00	00000000 FFFFFFFF FFFFFFFF FFFFFFFF 01000000 FFFF...	Ack	0538.435.166.654.12..	1111	1111
28607525	R→	MsgD (Vendor Defined Type)	16	00:00:00	FFFFFFFF 00000000 3A000100 01005000 39000000 5000...	Ack	0538.435.172.174.12..	1111	1111
28607527	R→	MsgD (Vendor Defined Type)	9	00:00:00	000000700 2F000000 FFFFFFFF FFFFFFFF FFFFFFFF...	Ack	0538.435.177.590.12..	1111	1111
28607529	R→	MsgD (Vendor Defined Type)	2	09:00:00	7E100043 00007300	Ack	0538.450.201.464.12..	1111	1111
28607531	R→	MsgD (Vendor Defined Type)	2	05:00:00	FE100044 00007398	Ack	0538.458.295.051.12..	1111	1111
28607533	R→	MsgD (Vendor Defined Type)	16	09:00:00	7E100020 01007486 08000000 90010000 0c000000 0000...	Ack	0538.476.348.720.12..	1111	1111
28607535	R→	MsgD (Vendor Defined Type)	16	09:00:00	03000000 38000000 03000000 00000000 03000000 4000...	Ack	0538.477.454.412.12..	1111	1111
28607537	R→	MsgD (Vendor Defined Type)	16	09:00:00	00000000 00000000 00000000 00000000 00000000 0000...	Ack	0538.477.949.634.12..	1111	1111
28607539	R→	MsgD (Vendor Defined Type)	10	09:00:00	00000000 00000000 00000000 00000000 00000000 0000...	Ack	0538.478.272.619.12..	1111	1111
28607541	R→	MsgD (Vendor Defined Type)	2	00:00:00	FE100044 01007486	Ack	0538.478.287.333.12..	1111	1111
28607543	R→	MsgD (Vendor Defined Type)	2	09:00:00	7E100040 01007400	Ack	0538.484.669.006.12..	1111	1111
28607545	R→	MsgD (Vendor Defined Type)	16	05:00:00	FE100020 01007400 90010000 00000000 09000000 0000...	Ack	0538.494.721.221.12..	1111	1111
28607547	R→	MsgD (Vendor Defined Type)	16	00:00:00	03000000 00000000 03000000 00000000 03000000 3800...	Ack	0538.494.727.081.12..	1111	1111
28607549	R→	MsgD (Vendor Defined Type)	16	00:00:00	00000000 00000000 00000000 00000000 00000000 3800...	Ack	0538.494.732.652.12..	1111	1111
28607551	R→	MsgD (Vendor Defined Type)	16	00:00:00	02000100 07000300 02010000 00000000 00000000 0000...	Ack	0538.494.738.140.12..	1111	1111
28607553	R→	MsgD (Vendor Defined Type)	16	00:00:00	05000000 00000101 02010000 10000000 00000000 0000...	Ack	0538.494.743.660.12..	1111	1111
28607555	R→	MsgD (Vendor Defined Type)	16	00:00:00	FFFFFF00 00000000 00000000 00000000 00000000 0000...	Ack	0538.494.749.102.12..	1111	1111
28607557	R→	MsgD (Vendor Defined Type)	9	05:00:00	FFFFFF0000 FFFF0000 00000000 00000000 00000000 0000...	Ack	0538.494.754.500.12..	1111	1111
28607559	R→	MsgD (Vendor Defined Type)	2	09:00:00	7E100043 00007400	Ack	0538.510.943.743.12..	1111	1111

DATA Flow显示了含有DATA 的封包信息。便于查看多个信号的DATA

Packet 28607513	R→	16.0 x8	TLP	1657	Msg	MsgID 011-10010	Msg Routing ID	Length 2	RequesterID 090.00.0	Tag 0	DeviceID 002.00.0	Message Code Vendor_Defined_Type1	VID DMTF	MCTP Header	Dst EID 0x0C	Src EID 0x09	MCTP Message Vendor Defined - PCI	Data FE 10 00 44 01 00 73 F3	LCRC 0xD85EDF8E	Time Delta 52.000 ns	Time Stamp 0538.417.486.283.120 s
Packet 28607514	R→	16.0 x8	DLLP	ACK	AckNak_Seq_Num 1657	CRC16 0x350F	Idle	17.619 ms	Time Stamp 0538.417.486.283.120 s												
Packet 28607515	R→	16.0 x8	TLP	1506	Msg	MsgID 011-10010	Msg Routing ID	Length 2	RequesterID 002.00.0	Tag 16	DeviceID 090.00.0	Message Code Vendor_Defined_Type1	VID DMTF	MCTP Header	Dst EID 0x09	Src EID 0x0C	MCTP Message Vendor Defined - PCI	Data 8 bytes	LCRC 0x13684F00	Time Delta 84.000 ns	Time Stamp 0538.435.105.623.120 s
Packet 28607516	R→	16.0 x8	DLLP	ACK	AckNak_Seq_Num 1506	CRC16 0xAD98	Idle	44.147 us	Time Stamp 0538.435.105.707.120 s												

QuickTiming markers not set

Data: Packet 28607513 (8 bytes)

Address

Hexadecimal

Decimal

00000000

FE 10

254 016

00000002

00 44

000 048

00000004

01 00

001 000

00000006

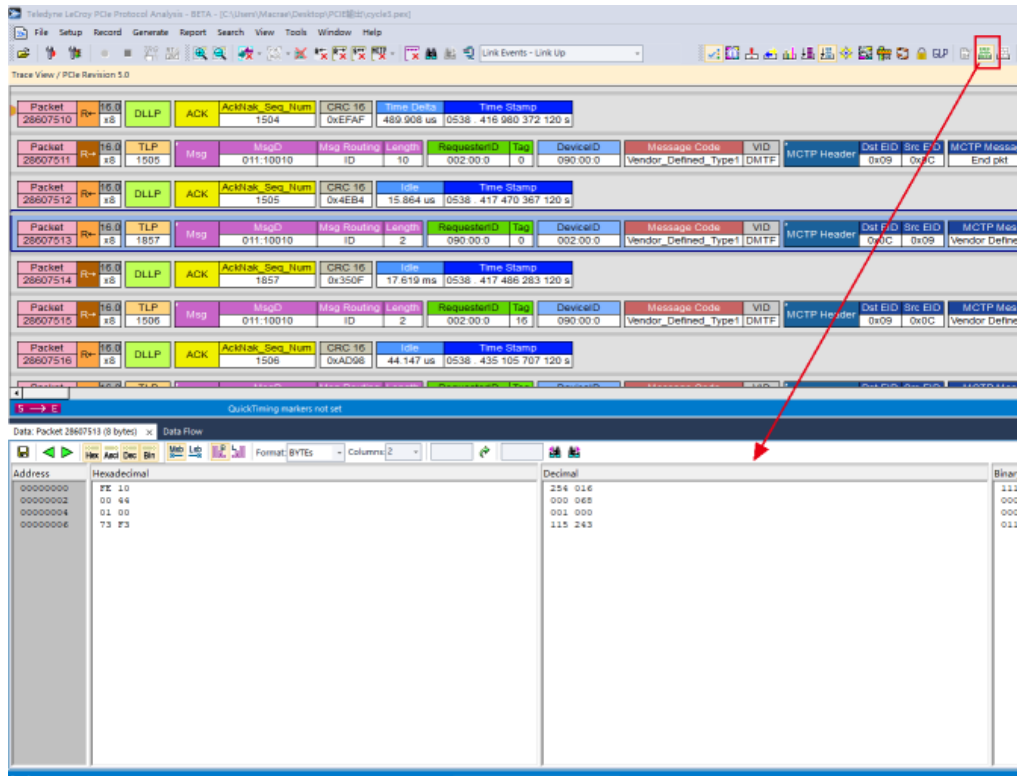
73 F3

115 243

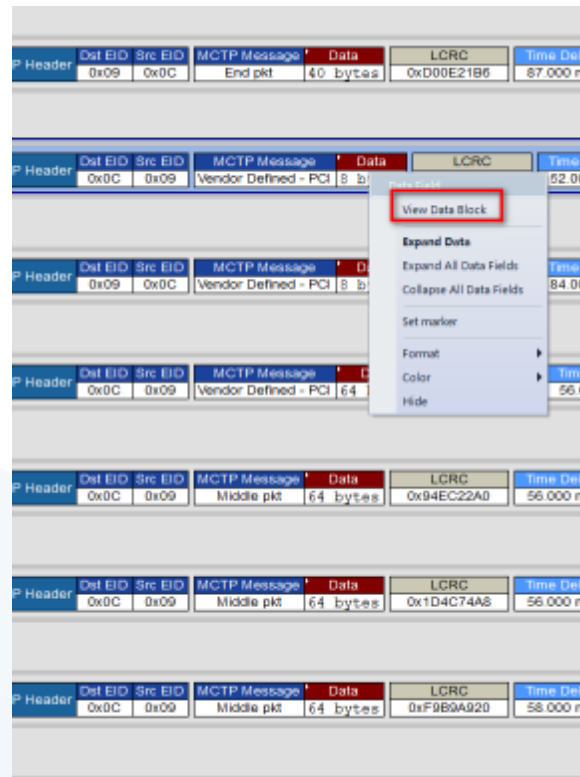
单独的DATA工具可以对DATA数据做更多的显示和处理。

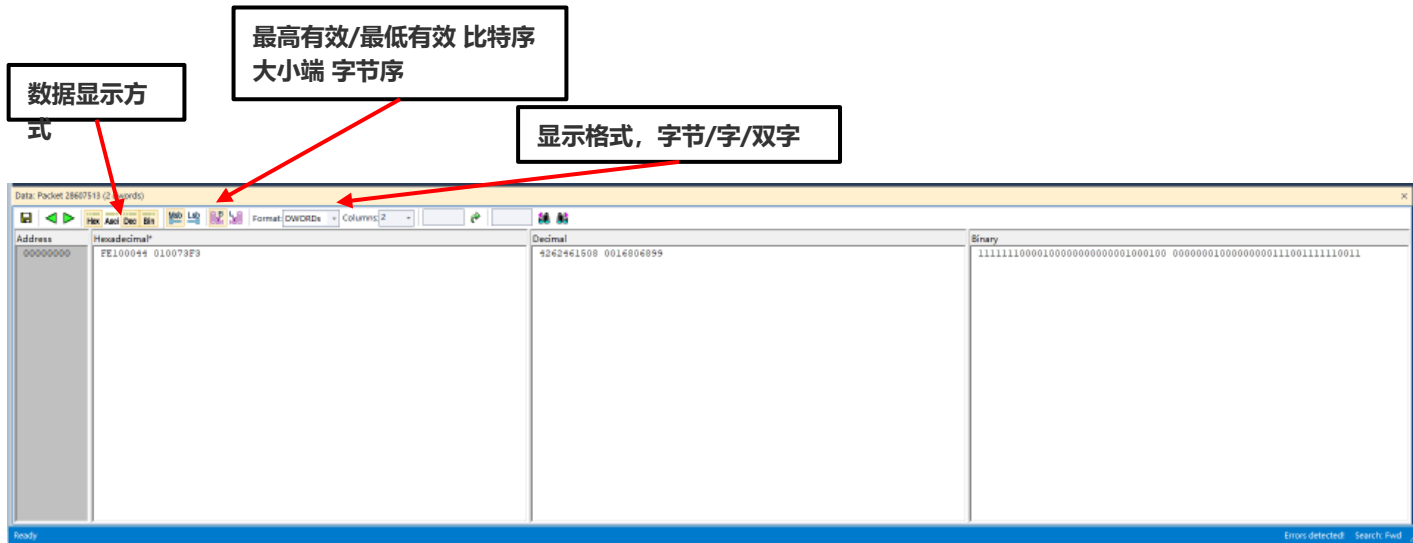
选中要看DATA的封包→Report→DATA

选中要看DATA的封包→点击工具栏的DATA



直接在想看的数据上方右键选择View Data Block





比特序:

MSB :最高有效位, 比特从高到低排列 (从左往右)

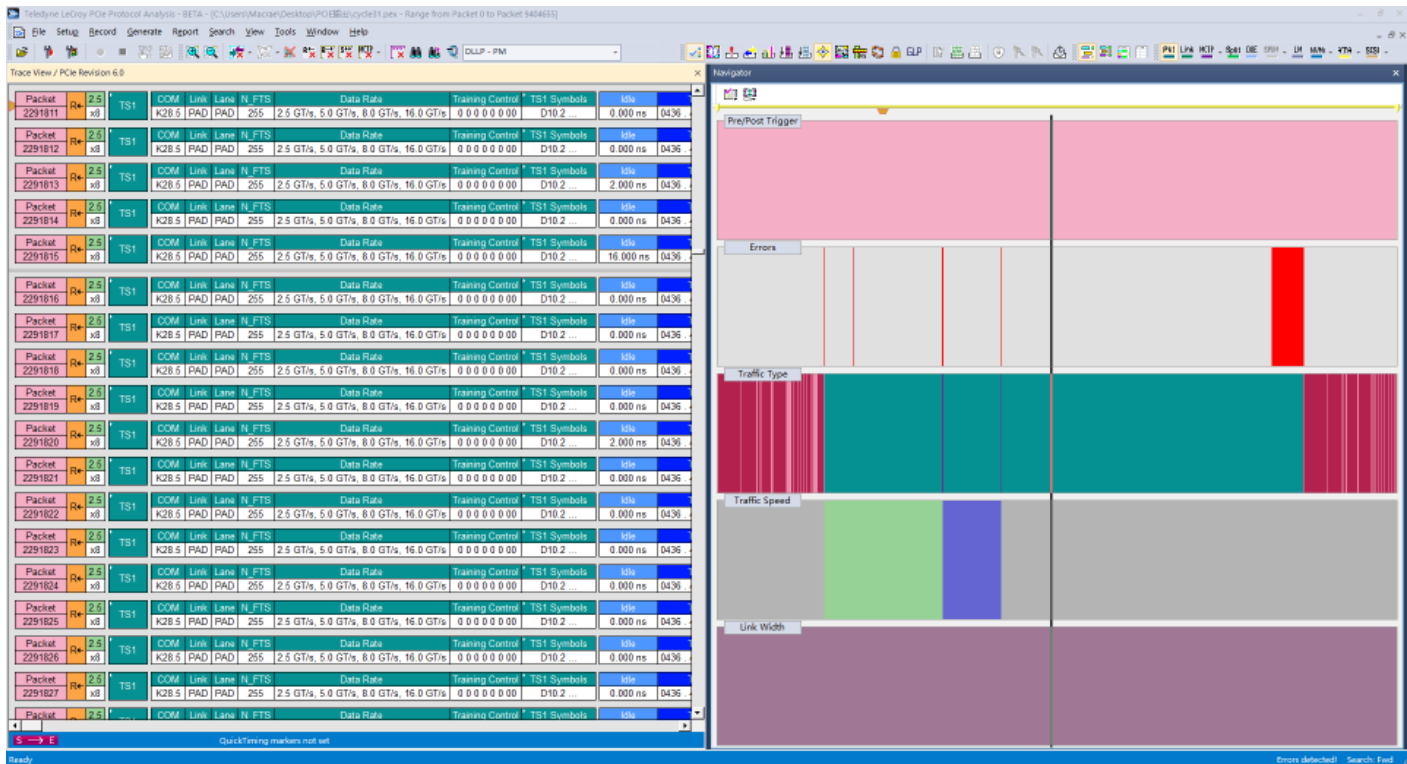
LSB :最低有效位, 比特从高到低排列 (从左往右)

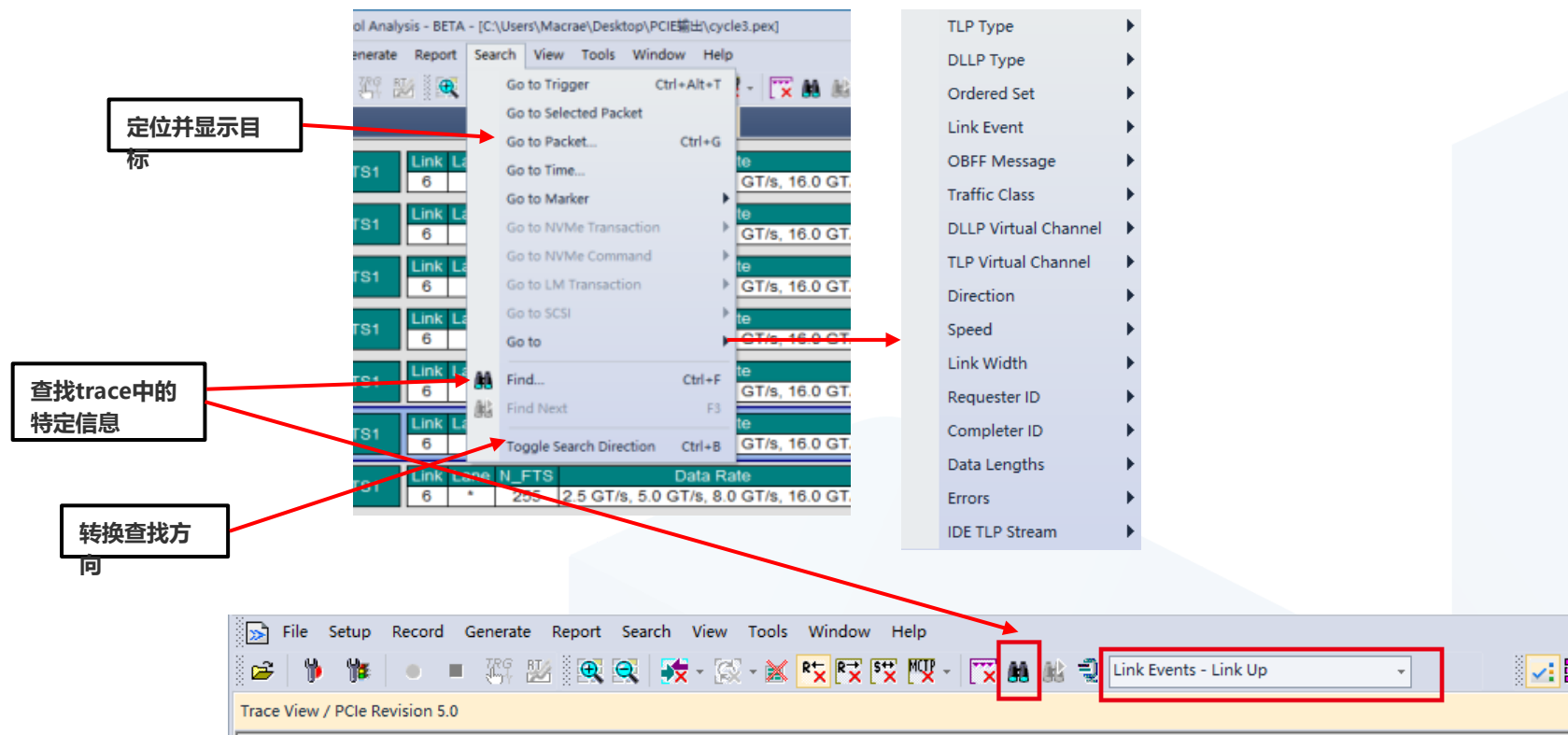
字节序:

Big endian :将高序**字节**存储在起始地址 (从左往右)

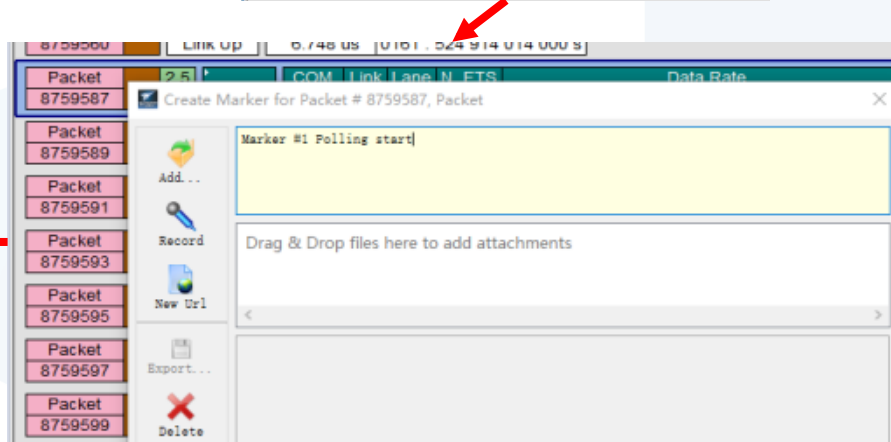
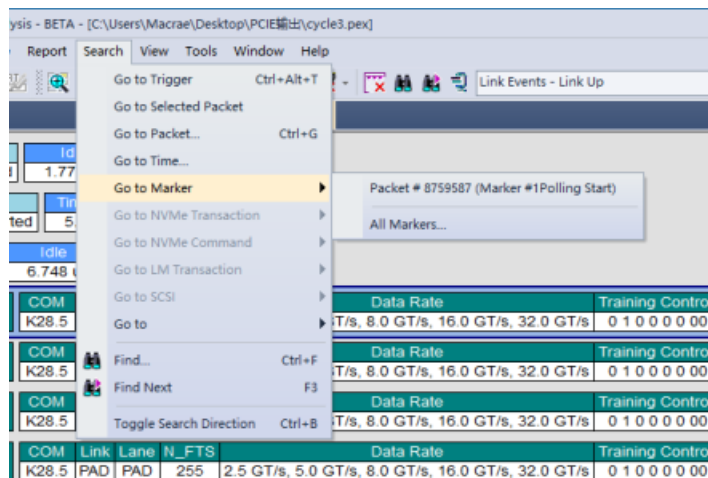
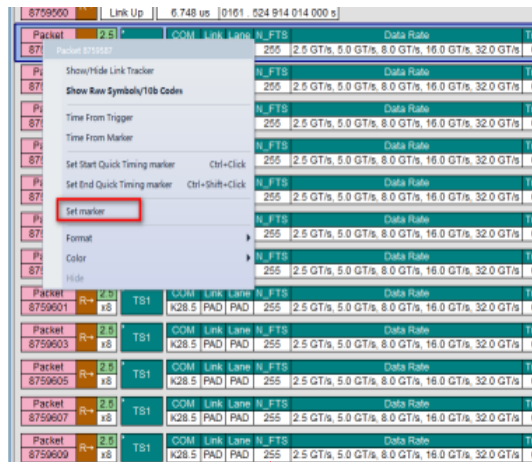
Little endian :将低序**字节**存储在起始地址 (从左往右)

Trace Navigator可以作为Trace的概览以及导航使用，较为快速查看trace的位宽，速率，error等变化点以及差异点。





在想标记的地方右键点击Set marker可以创建一个marker，然后点击GO TO Marker→目标marker就可以回到marker的位置



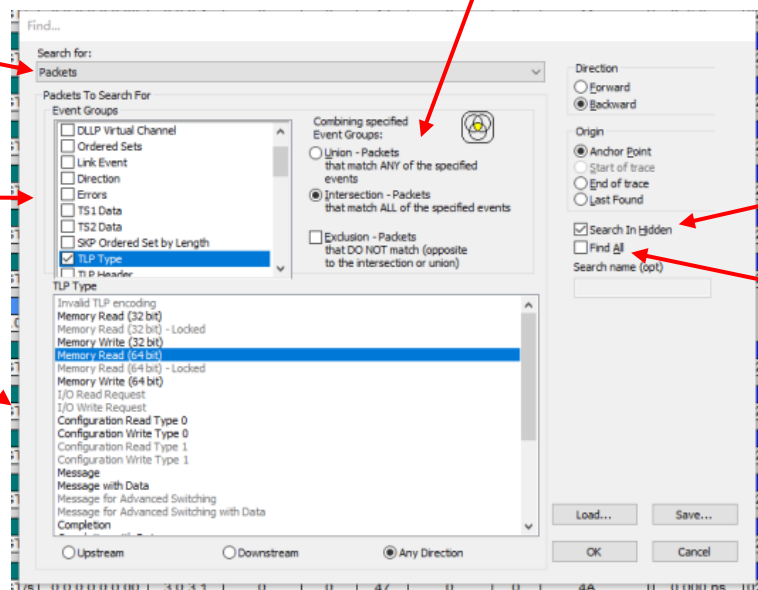
选择查找的类别，封包或者事务层信息等

设置查找条件，可以组合

对查找条件按照与或非进行查找

勾选就会把符合条件的隐藏目标也找到

勾选就会把所有符合条件的目标单独列出来，不勾选就一个个找



Search—Find

Teledyne LeCroy PCIe Protocol Analysis - ALPHA - [C:\Users\Public\Documents\LeCroy\PCIe Protocol Suite\Sample Files\Lane_Margining.pex]

File Setup Rec

Trace View

Find...

Search for: LM Transactions

LM Transactions To Search For

Event Groups

- ☒ Margin Command
- ☐ Margin Type
- ☐ Receiver Number
- ☐ Errors

Combining specified Event Groups:

- ☐ Union - LM Transactions that match ANY of the specified events
- ☒ Intersection - LM Transactions that match ALL of the specified events
- ☐ Exclusion - LM Transactions that DO NOT match (opposite to the intersection or union)

Margin Command

- No Command
- Access Retimer register
- Report Margin Control Capabilities
- Report M_NumVoltageSteps
- Report M_NumTimingSteps
- Report M_MaxTimingOffset
- Report M_SamplingRateVoltage
- Report M_SamplingRateTiming
- Report M_SampleCount
- Report M_MaxLanes
- Report Reserved
- Set Error Count Limit
- Go to Normal Settings

Direction

- ☒ Forward
- ☐ Backward

Origin

- ☒ Anchor Point
- ☐ Start of trace
- ☐ End of trace
- ☐ Last Found

☐ Find All

☒ Search In Hidden

Load... Save... OK Cancel

LM Transactions

LM	Command	Lane	MT	RN	MP	LM Response	Lane	MT	RN
13	R→	x4							
14	R→	x4							
15	R→	x4							
16	R→	x4							
17	R→	x4							
18	R→	x4							

MMMaxVoltageOffset 69

Time Delta 20.000 ms

Time Stamp 0000.350 003 400 0 s

MSamplingRateVoltage 18

Time Delta 20.000 ms

Time Stamp 0000.370 003 600 0 s

QuickTiming markers not set

View 菜单栏

Nettrix 宁畅

设置显示的菜单栏和工具栏

设置显示的菜单栏和工具栏

将trace中连续相同的部分压缩显示

按照封包显示

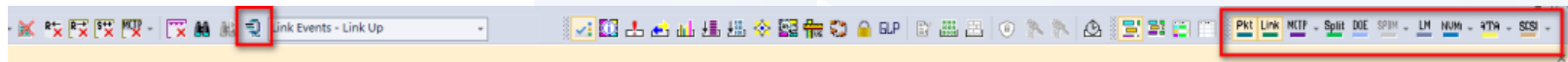
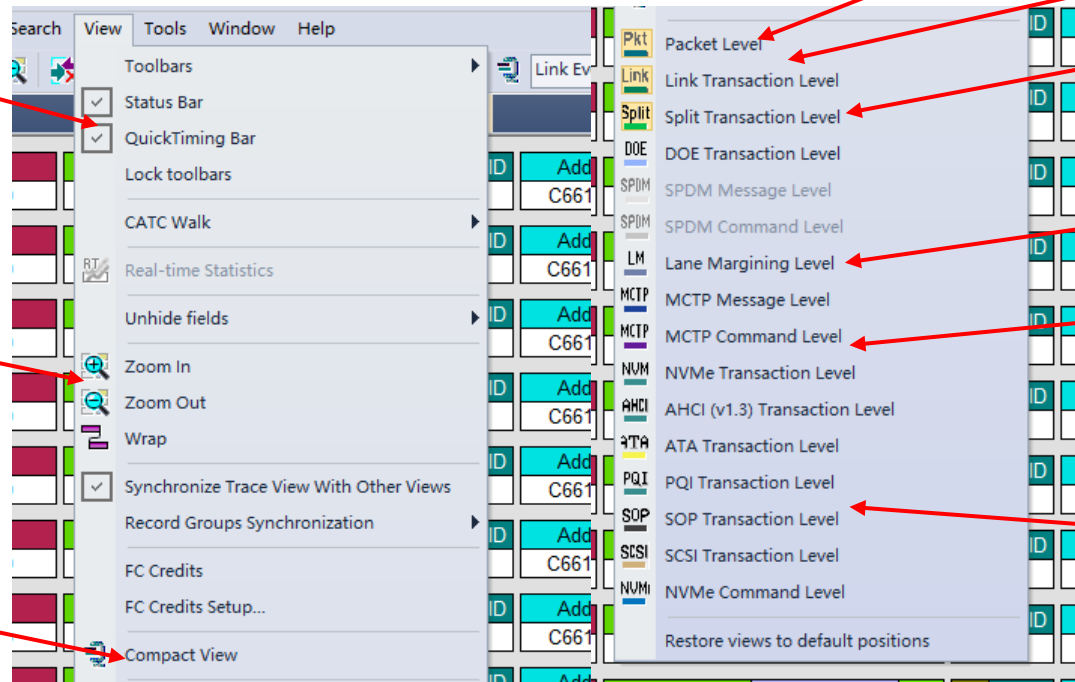
按照传输事务显示

按照分离事务显示

显示lane margin相关事务

内容
显示MCTP相关
相关事务内容

显示ATA NVM
SCSI相关事务内容



View — Compact View

Packet 1474632	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 152 000 s
Packet 1474633	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 216 000 s
Packet 1474634	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 280 000 s
Packet 1474635	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 344 000 s
Packet 1474636	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 408 000 s
Packet 1474637	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 472 000 s
Packet 1474638	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 536 000 s
Packet 1474639	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 600 000 s
Packet 1474640	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	2.000 ns	0428 456 463 664 000 s
Packet 1474641	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 730 000 s
Packet 1474642	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 794 000 s
Packet 1474643	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Idle	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	0.000 ns	0428 456 463 858 000 s

Packet 1474627	Link Event	Idle	Time Stamp									
	CLKREQ# Deasserted	1.114 sec	0427 335 289 882 000 s									
Packet 1474628	Link Event	Time Delta	Time Stamp									
	PERST# Deasserted	7.160 ms	0428 449 297 248 000 s									
Packet 1474629	Link Event	Idle	Time Stamp									
	Link Up	5.466 us	0428 456 457 556 000 s									
408484 TS Packets 1474639-1863113	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Time Delta	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	26.282 ms	0428 456 463 024 000 s
Packet 1863115	Link Event	Idle	Time Stamp									
	Link Down	15.645 ms	0428 482 755 310 000 s									
Packet 1863116	Link Event	Idle	Time Stamp									
	Link Up	5.470 us	0428 498 400 326 000 s									
408493 TS Packets 1863117-2291599	R+	2.5 x8	TS1	COM	Link	Lane	N_FTS	Data Rate	Training Control	TS1 Symbols	Time Delta	Time Stamp
				K28.5	PAD	PAD	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s, 16.0 GT/s	0 0 0 0 0 0 0	D10.2	6.243 sec	0428 498 405 796 000 s
Packet 2291600	Link Event	Time Delta	Time Stamp									
	DFCSTA Deasserted	87.816 us	0428 744 717 300 000 s									

以Packet为显示单元显示信息

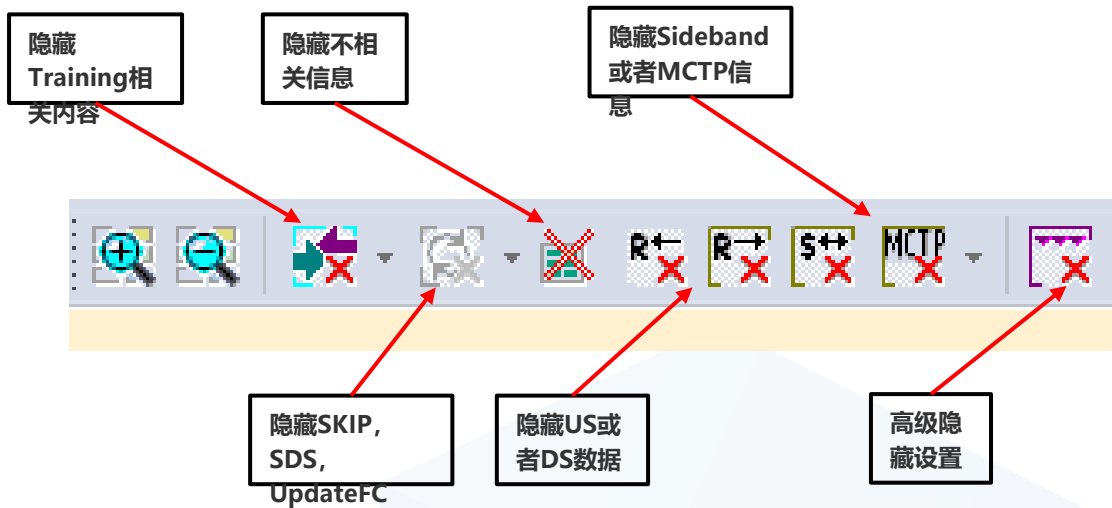
Packet 98	R+	16.0 x8	TLP	Msg	MsgD	Msg Routing	Length	RequesterID	Tag	Message Code	VID	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp
			26		011.10011	Broadcast	1	000.20.6	16	Vendor_Defined_Type1	DMTF		0xFF	0x08	Control	1 dword	0x199375AA	83.000 ns	0013 . 856 562 391 120 s
Packet 99	R+	16.0 x8	DLLP	ACK	AckNak_Seq_Num	CRC 16	Time Delta	Time Stamp											
					26	0xF892	1.009 sec	0013 . 856 562 474 120 s											
Packet 100	R+	16.0 x8	TLP	Msg	MsgD	Msg Routing	Length	RequesterID	Tag	Message Code	VID	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp
			27		011.10011	Broadcast	1	000.20.6	16	Vendor_Defined_Type1	DMTF		0xFF	0x08	Control	1 dword	0x0ED44970	85.000 ns	0014 . 865 883 958 120 s
Packet 101	R+	16.0 x8	DLLP	ACK	AckNak_Seq_Num	CRC 16	Time Delta	Time Stamp											
					27	0x5989	1.002 sec	0014 . 865 884 043 120 s											
Packet 102	R+	16.0 x8	TLP	Msg	MsgD	Msg Routing	Length	RequesterID	Tag	Message Code	VID	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp
			28		011.10011	Broadcast	1	000.20.6	16	Vendor_Defined_Type1	DMTF		0xFF	0x08	Control	1 dword	0xD1110C25	84.000 ns	0015 . 867 928 562 120 s
Packet 103	R+	16.0 x8	DLLP	ACK	AckNak_Seq_Num	CRC 16	Time Delta	Time Stamp											
					28	0x3ECB	1.008 sec	0015 . 867 928 646 120 s											
Packet 104	R+	16.0 x8	TLP	Msg	MsgD	Msg Routing	Length	RequesterID	Tag	Message Code	VID	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp
			29		011.10011	Broadcast	1	000.20.6	16	Vendor_Defined_Type1	DMTF		0xFF	0x08	Control	1 dword	0xDA58B90D	86.000 ns	0016 . 876 263 153 120 s
Packet 105	R+	16.0 x8	DLLP	ACK	AckNak_Seq_Num	CRC 16	Time Delta	Time Stamp											
					29	0x9FD0	1.002 sec	0016 . 876 263 239 120 s											
Packet 106	R+	16.0 x8	TLP	Msg	MsgD	Msg Routing	Length	RequesterID	Tag	Message Code	VID	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp
			30		011.10011	Broadcast	1	000.20.6	16	Vendor_Defined_Type1	DMTF		0xFF	0x08	Control	1 dword	0xBE99054A	84.000 ns	0017 . 878 259 288 120 s

PCIe一次事务包括TLP及与TLP对应的ACK (NAK) , Link Transaction Level就是按照一个个事务显示

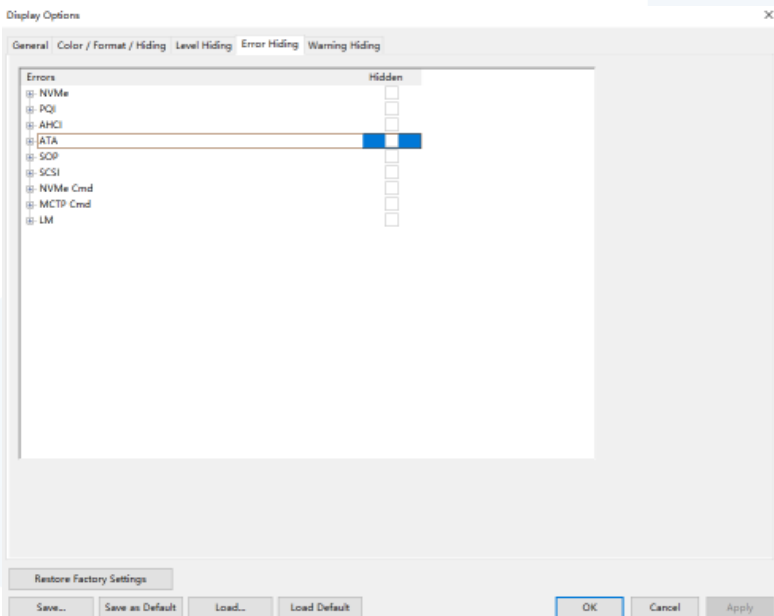
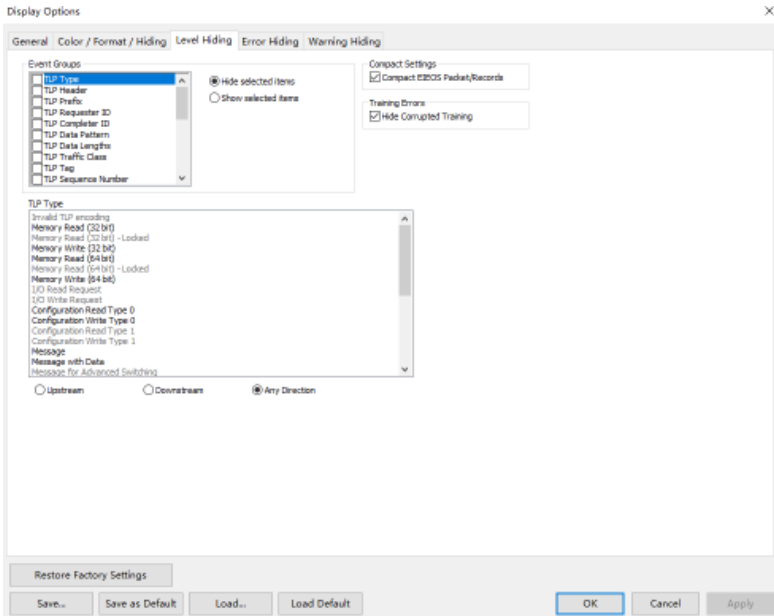
* Link Tra	47	R+	16.0	TLP	Cty	CtyID	Length	RequesterID	Tag	DeviceID	Register	1st BE	VC-ID	Explicit ACK	Metrics	#Packets	Time Delta	Time Stamp					
			x8	25		000:00100	1	252:00:2	0	090:00:0	0x074	1111	0	Packet #95	2	2	4.093 us	0004 . 418 943 714 120 s					
* Link Tra	48	R+	16.0	TLP	Cpl	CplD	Length	RequesterID	Tag	CompleterID	Status	BCM	Byte Cnt	Lwr Addr	Register Data	VC-ID	Explicit ACK	Metrics	#Packets	Time Delta	Time Stamp		
			x8	23		010:01010	1	252:00:2	0	000:00:0	SC	0	4	0x00	0x0042E884	0	Packet #97	2	2	9.438 sec	0004 . 418 947 807 120 s		
* Link Tra	49	R+	16.0	TLP	Msg	MsgID	Msg Routing	Length	RequesterID	Tag	Message Code	VD	MCTP Header	Dst EID	Src EID	MCTP Message	Data	VC-ID	Explicit ACK	Metrics	#Packets	Time Delta	Time Stamp
			x8	26		011:10011	Broadcast	1	000:20:6	16	Vendor_Defined_Type1	DMTF	0xFF	0x08	Control	1 dword	0	Packet #99	2	2	83.000 ns	0013 . 856 562 391 120 s	
Packet	98	R+	16.0	TLP	Msg	MsgID	Msg Routing	Length	RequesterID	Tag	Message Code	VD	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp			
			x8	26		011:10011	Broadcast	1	000:20:6	16	Vendor_Defined_Type1	DMTF	0xFF	0x08	Control	1 dword	0x199375AA	83.000 ns	0013 . 856 562 391 120 s				
Packet	99	R+	16.0	DLLP	ACK	AckNak_Seq Num	CRC-16	Time Delta	Time Stamp														
			x8		26		0xF892	1.009 sec	0013 . 856 562 474 120 s														
* Link Tra	50	R+	16.0	TLP	Msg	MsgID	Msg Routing	Length	RequesterID	Tag	Message Code	VD	MCTP Header	Dst EID	Src EID	MCTP Message	Data	VC-ID	Explicit ACK	Metrics	#Packets	Time Delta	Time Stamp
			x8	27		011:10011	Broadcast	1	000:20:6	16	Vendor_Defined_Type1	DMTF	0xFF	0x08	Control	1 dword	0	Packet #101	2	2	85.000 ns	0014 . 865 883 958 120 s	
Packet	100	R+	16.0	TLP	Msg	MsgID	Msg Routing	Length	RequesterID	Tag	Message Code	VD	MCTP Header	Dst EID	Src EID	MCTP Message	Data	LCRC	Time Delta	Time Stamp			
			x8	27		011:10011	Broadcast	1	000:20:6	16	Vendor_Defined_Type1	DMTF	0xFF	0x08	Control	1 dword	0x0ED44970	85.000 ns	0014 . 865 883 958 120 s				
Packet	101	R+	16.0	DLLP	ACK	AckNak_Seq Num	CRC-16	Time Delta	Time Stamp														
			x8		27		0x5989	1.002 sec	0014 . 865 884 043 120 s														
* Link Tra	51	R+	16.0	TLP	Msg	MsgID	Msg Routing	Length	RequesterID	Tag	Message Code	VD	MCTP Header	Dst EID	Src EID	MCTP Message	Data	VC-ID	Explicit ACK	Metrics	#Packets	Time Delta	Time Stamp
			x8	28		011:10011	Broadcast	1	000:20:6	16	Vendor_Defined_Type1	DMTF	0xFF	0x08	Control	1 dword	0	Packet #103	2	2	1.008 sec	0015 . 867 928 562 120 s	
* Link Tra	52	R+	16.0	TLP	Msg	MsgID	Msg Routing	Length	RequesterID	Tag	Message Code	VD	MCTP Header	Dst EID	Src EID	MCTP Message	Data	VC-ID	Explicit ACK	Metrics	#Packets	Time Delta	Time Stamp
			x8	29		011:10011	Broadcast	1	000:20:6	16	Vendor_Defined_Type1	DMTF	0xFF	0x08	Control	1 dword	0	Packet #105	2	2	1.002 sec	0016 . 876 263 153 120 s	

对于需要回复消息的TLP，PCIe允许回复TLP不必和请求TLP相连，这种分离的事务处理方式叫做分离事务。Split Transaction Level就是将分离事务中的TLP放在一起显示。

Split Tra	R→	16.0	Mem	MRd(32)	RequesterID	CompleterID	Tag	TC	VC ID	Address	Status	Data	Metrics	# LinkTrans	Time Delta	Time Stamp				
190082		x8		000.00000	253.00.2	090.00.0	256	0	0	C6619BBC	SC	1 dword		2	596.000 ns	0477 . 810 862 098 120 s				
Split Tra	R→	16.0	Mem	MRd(32)	RequesterID	CompleterID	Tag	TC	VC ID	Address	Status	Data	Metrics	# LinkTrans	Time Delta	Time Stamp				
190083		x8		000.00000	253.00.2	090.00.0	256	0	0	C6619BC0	SC	1 dword		2	590.000 ns	0477 . 810 862 694 120 s				
Split Tra	R→	16.0	Mem	MRd(32)	RequesterID	CompleterID	Tag	TC	VC ID	Address	Status	Data	Metrics	# LinkTrans	Time Delta	Time Stamp				
190084		x8		000.00000	253.00.2	090.00.0	256	0	0	C6619BC4	SC	1 dword		2	588.000 ns	0477 . 810 863 284 120 s				
Split Tra	R→	16.0	Mem	MRd(32)	RequesterID	CompleterID	Tag	TC	VC ID	Address	Status	Data	Metrics	# LinkTrans	Time Delta	Time Stamp				
190085		x8		000.00000	253.00.2	090.00.0	256	0	0	C6619BC8	SC	1 dword		2	80.000 ns	0477 . 810 863 872 120 s				
Link Tra	R→	16.0	TLP	Mem	MRd(32)	Length	RequesterID	Tag	Address	1st BE	Last BE	VC ID	Explicit ACK	Metrics	# Packets	Time Delta	Time Stamp			
791148		x8	2363		000.00000	1	253.00.2	256	C6619BC8	1111	0000	0	Packet #8218302		2	80.000 ns	0477 . 810 863 872 120 s			
Packet	R→	16.0	TLP	Mem	MRd(32)	Length	RequesterID	Tag	Address	1st BE	Last BE	LCRC	Time Delta	Time Stamp						
8218301		x8	2363		000.00000	1	253.00.2	256	C6619BC8	1111	0000	0xF0AF7D5E	80.000 ns	0477 . 810 863 872 120 s						
Packet	R→	16.0	DLLP	ACK	AckNak_Seq_Num	CRC 16	Time Delta	Time Stamp												
8218302		x8		2363	0x3E9F	134.000 ns	0477 . 810 863 952 120 s													
Link Tra	R→	16.0	TLP	Cpl	CplID	Length	RequesterID	Tag	CompleterID	Status	BCM	Byte Cnt	Lwr Addr	Data	VC ID	Explicit ACK	Metrics	# Packets	Time Delta	Time Stamp
791149		x8	2355		010.01010	1	253.00.2	256	090.00.0	SC	0	4	0x48	1 dword	0	Packet #8218304		2	57.000 ns	0477 . 810 864 086 120 s
Packet	R→	16.0	TLP	Cpl	CplID	Length	RequesterID	Tag	CompleterID	Status	BCM	Byte Cnt	Lwr Addr	Data	LCRC	Time Delta	Time Stamp			
8218303		x8	2355		010.01010	1	253.00.2	256	090.00.0	SC	0	4	0x48	1 dword	0x9BC62D1B	57.000 ns	0477 . 810 864 086 120 s			
Packet	R→	16.0	DLLP	ACK	AckNak_Seq_Num	CRC 16	Time Delta	Time Stamp												
8218304		x8		2355	0x3642	325.000 ns	0477 . 810 864 143 120 s													
Split Tra	R→	16.0	Mem	MRd(32)	RequesterID	CompleterID	Tag	TC	VC ID	Address	Status	Data	Metrics	# LinkTrans	Time Delta	Time Stamp				
190086		x8		000.00000	253.00.2	090.00.0	256	0	0	C6619BCC	SC	1 dword		2	591.000 ns	0477 . 810 864 468 120 s				
Split Tra	R→	16.0	Mem	MRd(32)	RequesterID	CompleterID	Tag	TC	VC ID	Address	Status	Data	Metrics	# LinkTrans	Time Delta	Time Stamp				
190087		x8		000.00000	253.00.2	090.00.0	256	0	0	C6619BD0	SC	1 dword		2	588.000 ns	0477 . 810 865 059 120 s				

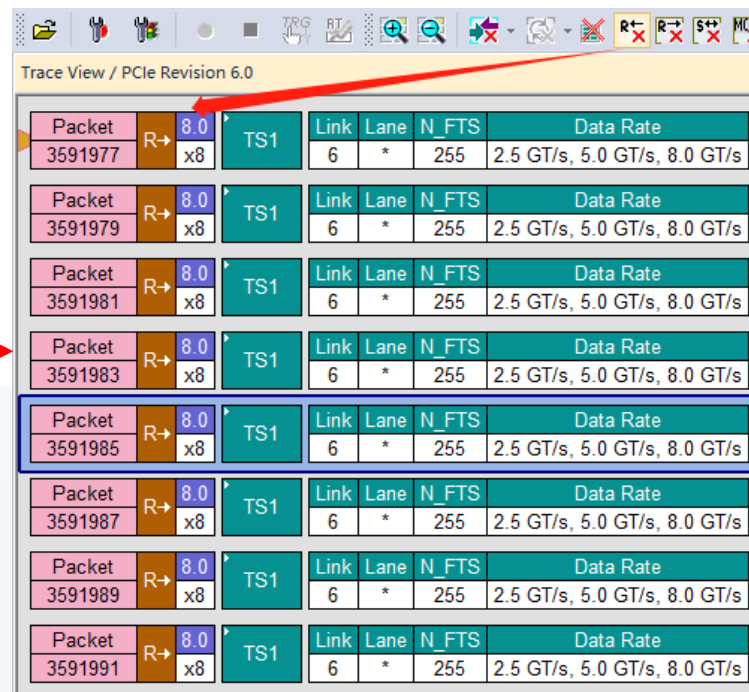


通过配置隐藏的参数，可以将不关心的部分隐藏，在合适的部分通过显示Error也可以定位问题。



隐藏US或者DS可以方便查看单一方向的trace。下图便是隐藏US后效果。

Packet				Link	Lane	N_FTS	Data Rate
3591977	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591978	R←	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591979	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591980	R←	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591981	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591982	R←	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591983	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591984	R←	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591985	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s



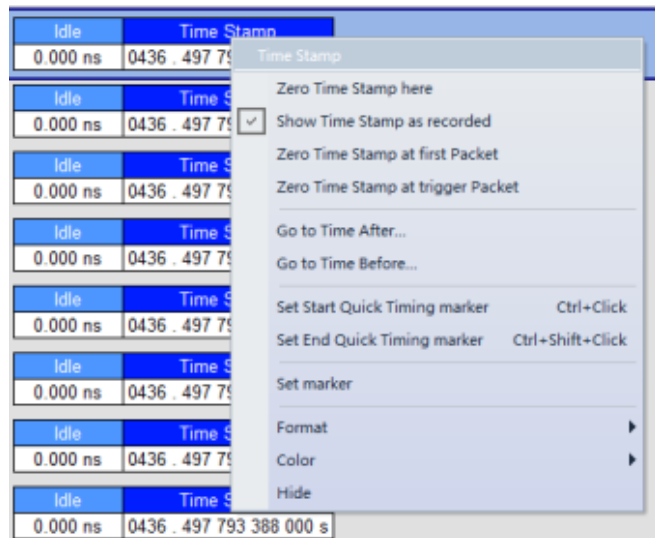
Packet				Link	Lane	N_FTS	Data Rate
3591977	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591979	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591981	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591983	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591985	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591987	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591989	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s
3591991	R→	8.0	TS1	6	*	255	2.5 GT/s, 5.0 GT/s, 8.0 GT/s

Time Delta和Time Idle是两个Package之间时间的描述方式。

Object	TS1 Symbols	Time Delta	Time Stamp
0	4A ...	4.880 ns	0442 . 315 914 039 240 s
Object	TS1 Symbols	This time delta is measured from the beginning of this Packet to the beginning of the next Packet on the screen	
0	4A ...	3.000 ns	0442 . 315 914 044 120 s

TS1 Symbols	Idle	Time Stamp
D10.2 ...	0.000 ns	0436 . 497 793 580 000 s
Idle Time between the end of this Packet and start of the next one (+/- 8 nSec)		
TS1 Symbols	Idle	Time Stamp
D10.2 ...	0.000 ns	0436 . 497 793 644 000 s

在Time Stamp上右键可以选择设置0s的时间起始点。



Time Calculator

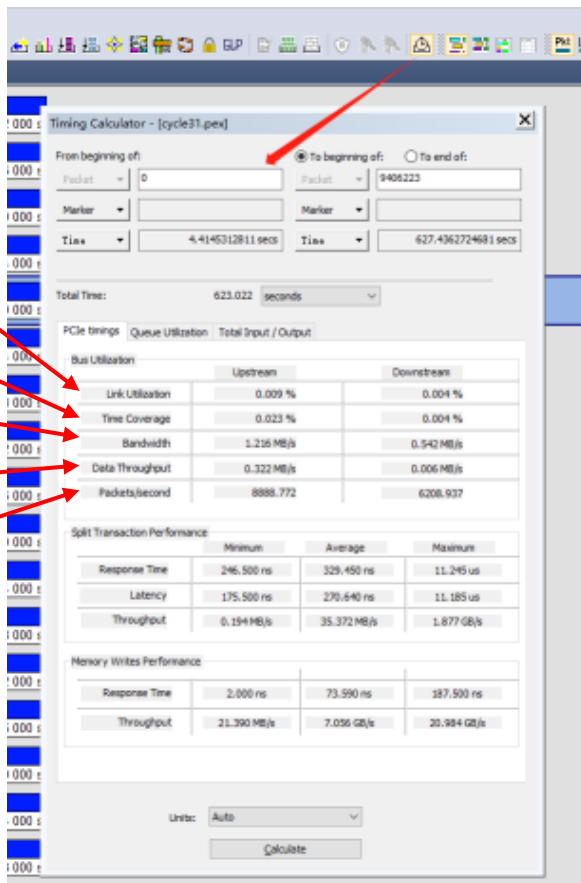
链路利用率: $\frac{\text{packet symbol}}{\text{total symbol}}$

时间覆盖

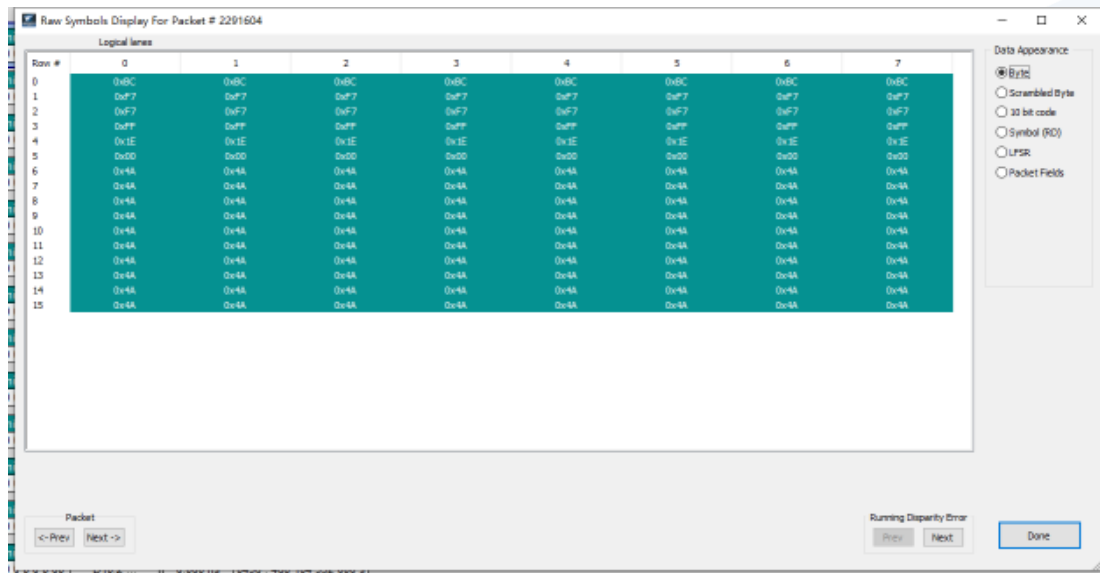
信号带宽: $\frac{\text{信号速}}{\text{率}}$

数据带宽: $\frac{\text{时间内的数据平均速率}}{\text{率}}$

每秒传输的
packet



在Packet数字段上面右键选择Show Row Symbol，或者双击Packet数字段就可以进入Raw symbol显示框。



Row Symbol

Raw Symbols Display For Packet # 2291604

Logical lanes

Row #	0	1	2	3	4	5	6	7
0	0x8C	0x8C	0x8C	0x8C	0x8C	0x8C	0x8C	0x8C
1	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7
2	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7
3	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7	0xF7
4	0x1E	0x1E	0x1E	0x1E	0x1E	0x1E	0x1E	0x1E
5	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00
6	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
7	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
8	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
9	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
10	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
11	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
12	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
13	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
14	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A
15	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A	0x4A

Packet: <- Prev Next ->

Running Disparity Error: Prev Next Done

显示字节信息（去扰后的）

显示去扰前的数据

显示8b10b编码后的数据（仅限 Gen1/Gen2）

显示8b10b编码前的数据（仅限 Gen1/Gen2）

显示LFSR

Seed

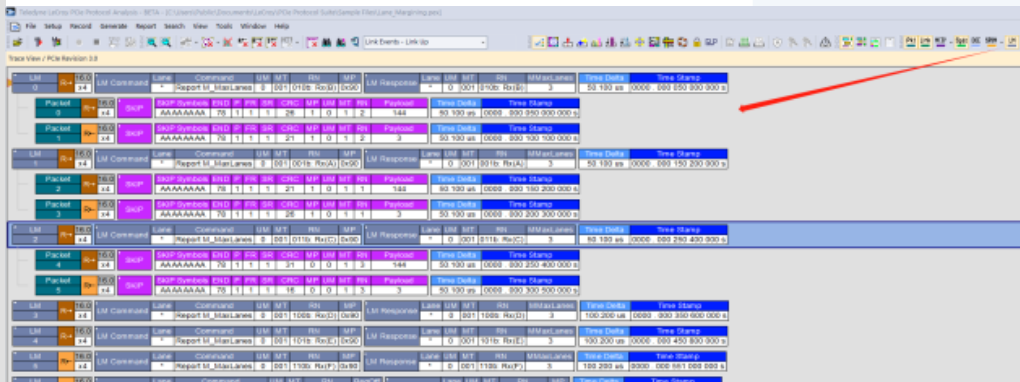
显示解析后的字段

对于所有支持数据速率为16.0 GT/s或更高的端口，包括伪端口(retimer)，都必须在接收端支持Lane margin。 Lane margin的命令和回复通过Control SKP Order Set来传输。

Table 4-53 Control SKP Ordered Set with 128b/130b Encoding

Symbol Number	Value	Description
0 through (4*N-1) [N can be 1 through 5]	AAh for 8.0 GT/s and 16.0 GT/s data rates 99h for 32.0 GT/s and higher data rates	SKP Symbol. Symbol 0 is the SKP Ordered Set identifier.
4*N	78h	SKP_END_CTL Symbol. Signifies the end of the Control SKP Ordered Set after three more Symbols.
4*N + 1	00-FFh	Bit 7: Data Parity Bit 6: First Retimer Data Parity Bit 5: Second Retimer Parity Bits [4:0]: Margin CRC [4:0]
4*N + 2	00-FFh	Bit 7: Margin Parity Bits [6:0]: Refer to § Section 4.2.18.1
4*N + 3	00-FFh	Bits [7:0]: Refer to § Section 4.2.18.1

Packet	R	x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
Packet 6	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAA	78	1	1	1	4	1	0	1	4	144	50.100 us	0000_000_350 600 000 s
Packet 7	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAA	78	1	1	1	11	1	0	1	4	3	50.100 us	0000_000_400 700 000 s
Packet 8	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAA	78	1	1	1	1	0	0	1	5	144	50.100 us	0000_000_450 800 000 s
Packet 9	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Idle	Time Stamp
				AAAAAAAA	78	1	1	1	14	0	0	1	5	3	50.096 us	0000_000_500 900 000 s
Packet 10	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAA	78	1	1	1	14	0	0	1	6	144	50.100 us	0000_000_551 000 000 s
Packet 11	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAA	78	1	1	1	1	0	0	1	6	3	50.100 us	0000_000_601 100 000 s
Packet 12	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAAAA	78	1	1	1	4	0	0	1	2	64	50.100 us	0000_000_651 200 000 s
Packet 13	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Idle	Time Stamp
				AAAAAAAAAA	78	1	1	1	7	1	0	1	2	0	50.094 us	0000_000_701 300 000 s
Packet 14	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAAAA	78	1	1	1	26	0	0	1	4	64	50.100 us	0000_000_751 400 000 s
Packet 15	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAAAA	78	1	1	1	10	0	0	1	4	5	50.100 us	0000_000_801 500 000 s
Packet 16	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAAAA	78	1	1	1	30	0	0	1	2	136	50.100 us	0000_000_851 600 000 s
Packet 17	R	10.0 x4	SKP	SKP Symbols	END	P	FR	SR	CRC	MP	UM	MT	RN	Payload	Time Delta	Time Stamp
				AAAAAAAAAA	78	1	1	1	17	0	0	1	2	27	50.100 us	0000_000_901 700 000 s



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